Earnings-Call and Filing NLP Signal Engine

Do changes in the language of SEC filings predict stock returns? A tested answer: no significant effect with a word-list baseline.

What I built

- **Pipeline (Python, SQLite):** downloads 10-K/10-Q MD&A sections and 8-K earnings releases from SEC EDGAR, scores six language features per document (guidance direction, hedging, tone, supply-chain / FX / demand risk), and caches results.
- **Signal:** quarter-over-quarter change in each feature, standardised within document type and calendar quarter. Feature signs were fixed *before* looking at returns, so nothing is fitted to the data.
- **Event study:** market-model abnormal returns vs SPY at 1, 5 and 20 trading days from the first tradable session after the SEC acceptance time (reaction day skipped in the main runs).
- **Tests:** quintile long-short spreads, rank information coefficient (IC), regression with errors clustered by quarter, out-of-sample split at 2022-01-01, 10 bps per-leg costs.
- **Engineering:** 12 automated tests; offline synthetic demo with a planted effect that the pipeline recovers, and finds nothing when the effect is set to zero.

Data

30 large-cap US stocks, filings from 2020: 1,036 events (667 from 10-K/10-Q, 369 from earnings releases).

Results

Run	Events	5d IC (t)	20d IC (t)	20d long-short (t)	20d OOS IC
All filings, skip reaction day	1,036	0.03 (0.86)	0.03 (0.74)	0.57% (0.85)	-0.01
All filings, include reaction day	1,036	0.04 (0.89)	0.05 (1.35)	1.15% (1.59)	0.01
Earnings releases only	369	0.02 (0.31)	0.05 (0.89)	n/a	0.00
10-K / 10-Q only	667	0.04 (0.97)	0.02 (0.41)	0.48% (0.67)	-0.01

IC = rank correlation of signal with abnormal return, averaged over quarters; t-statistics are across quarters. Long-short is n/a for releases: too few events per quarter to form five groups.

Conclusion

- No t-statistic exceeds 2 in absolute value in any run. Signs flip across horizons and subsets, and out-of-sample IC is about zero.
- Including the reaction day roughly doubles the 20-day spread (it adds the untradable part of the move) but is still not significant. Splitting by document type revealed nothing.
- This is a finding about the design (crude text measure, small sample of efficient stocks), not proof that filing language never matters.

Limitations

- About 1,000 events: low power, small effects not ruled out. Earnings releases found by a filename heuristic, so 369 is probably an undercount.
- Survivorship bias (today's large caps); 10-K/10-Q often follow the earnings news; word lists ignore context; flat cost assumption; no transcripts (no free source).
- An LLM extractor is built but was not run, so whether it beats the word-list baseline is untested.

Resume line

Built an end-to-end NLP event-study pipeline (SEC EDGAR, Python, SQLite, market-model abnormal returns) testing whether changes in filing language predict stock returns across 30 large caps (~1,000 filings, 2020 onward); found no significant effect, with robustness checks across horizons, document types, execution timing and an out-of-sample split.